LESSON PLAN

A/C ENGINEERING SYSTEM (17)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	03
Total No. of Lesson Plans	30	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Fuel, major classification of fuel based on fuel state, turbine engine fuels
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Fuel, major classification of fuel based on fuel state, turbine engine fuels
5. Specific Objectives	17.1.1.1 Trainee will be able to understand the construction and requirements of typical aircraft fuel system and its components.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of fuel system diagrams in IB; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the aircraft fuel system?
(b) Name two common types of fuel used in aviation.
(c) Why is fuel quality critical for turbine engines?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Liquid vs Gaseous fuel states • Jet A and Jet A-1 specifications • Kerosene-based fuel properties • Fuel density and freezing points • Contamination prevention methods	L & Ds	PPT, IB	15 min
(b)	• Turbine engine fuel requirements • Flash point and volatility limits • Additives for corrosion inhibition • Water separation mechanisms • Fuel filtration standards	L & Ds	PPT, PDF	20 min
(c)	• Fuel tank construction basics • Ventilation and pressure systems • Fuel pump operation principles • Fuel flow measurement devices • Emergency fuel shutoff procedures	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between Jet A and Jet A-1 fuels.
(b) Describe how turbine engines handle fuel contamination.
(c) List three key components of a typical aircraft fuel system.

12. Home Assignment and Follow Up	Prepare a brief note on the differences between reciprocating engine fuels and turbine engine fuels, referencing LP 02 concepts.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Brief description and purpose of inverted flight valve and recuperate.	Refer to the master syllabus reference list	LP 04

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Lieutenant, P No. 12345	Senior Instructor, Commander, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min